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One Edu — Hosting & Infrastructure Architecture

Cloud Hosting Design

Document	Hosting & Infrastructure Architecture
Product	One Edu — Education Super App
Prepared for	Board of Directors
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Cloud platform	Microsoft Azure
Primary region	UK South (UK West for DR)
Status	For review

Document Revision History

Version	Date	Author	Summary of change
0.9	June 2026	Architecture Team	Internal draft for review.
1.0	June 2026	Architecture Team	Initial board release — single-region & multi-region designs, component briefs, security & governance briefing.
1.1	June 2026	Architecture Team	Identity briefs + diagrams updated with social providers (Google, Apple, Microsoft/Outlook, Facebook).

1. Executive Summary

One Edu will be hosted on **Microsoft Azure** as a secure, compliant, multi-tenant SaaS platform. The design is deliberately **phased** to control cost and risk:

- **Launch (Phase 1):** a single Azure region hosting the first institution, built on managed **Azure App Service** and **Azure SQL Database** — a lean, production-grade footprint with high availability.
- **Scale (future):** the *same* blueprint is replicated into additional geographic regions, giving each country its own data boundary. This proves the platform is **expansion-ready** without re-architecture.

Two architectures are presented separately in this document:

1. **Single-Region (Single-Tenant) Architecture** — the day-one production environment (Section 5).
2. **Multi-Region (Multi-Tenant) Architecture** — the target operating model as the platform grows internationally (Section 6).

The estimated monthly hosting + tooling cost (excluding staff) for the launch environment is summarised in Section 9 and detailed in the companion workbook *CostEstimate-OneEdu-Board.xlsx*.

2. Purpose & Scope

Purpose. To give the Board a clear, non-exhaustive view of how One Edu is hosted, why each major component exists, and how the design satisfies security, data-protection, availability and growth requirements.

In scope: cloud hosting topology, core platform services, environments, security & compliance posture, high-availability & disaster recovery, and the expansion model.

Out of scope: application feature design (covered by the SRS), detailed network IP plans, and day-rate / staffing costs.

3. Architecture Principles

#	Principle	What it means in practice
P1	Managed-first	Prefer Azure platform (PaaS) services over self-managed servers — less to patch, operate and secure.
P2	Secure & compliant by design	Tenant isolation, encryption,

#	Principle	What it means in practice
		and data-residency are built in from day one, not added later.
P3	Phased investment	Pay for capability as modules go live; avoid over-provisioning at launch.
P4	Expansion-ready	One regional blueprint, stamped per country — growth needs no redesign.
P5	High availability	No single point of failure for production; automated recovery.
P6	One core technology set	Standardise on Azure App Service + Azure SQL to keep the team's operating skills focused.

4. Solution Overview

One Edu is delivered through **native mobile apps (iOS/Android)** and a **responsive web app**, all served from a hardened Azure environment. Traffic enters through a global edge with a Web Application Firewall, is processed by the application tier, and reads/writes a securely isolated data tier. Supporting services provide identity, notifications, AI, monitoring and security.

The platform is organised into a **Phase 1 (launch)** core and a **Phase 2 (growth)** set of services that are switched on as the corresponding product modules are released.

5. Single-Region (Single-Tenant) Architecture

This is the **day-one production environment** — one Azure region hosting the first institution (~5,000 students).

5.1 How it works (request flow)

1. Users on the **mobile app** or **web app** connect over HTTPS to **Azure Front Door**, which terminates TLS and applies the **Web Application Firewall**.
2. Front Door routes to **Azure App Service**, which runs the application (server-rendered UI + API).
3. The app authenticates users via **Microsoft Entra External ID** (sign-in + multi-factor authentication).
4. The app reads and writes the **data tier** over private connections: **Azure SQL** (tenant data, with Row-Level Security), **Blob Storage** (files + immutable audit log) and **Key Vault** (secrets and encryption keys).
5. **Communication Services** sends email/SMS/OTP; **App Insights / Log Analytics** and **Defender for Cloud** provide monitoring and threat detection.
6. As modules launch, the **Phase 2** services (real-time messaging, caching, async jobs, push notifications, AI) are enabled and the app pushes updates back to devices (push via APNS/FCM, live updates via WebSocket).

5.2 Component briefs — Phase 1 (launch)

Component	Azure service	Role / why it's here
Edge & firewall	Azure Front Door + WAF	Single secure entry point; TLS, global routing, blocks common web attacks (OWASP).
Application tier	Azure App Service (Premium v3)	Runs the web/API workload; auto-scales; zone-redundant for HA.
Identity	Microsoft Entra External ID	Sign-in with social logins (Google, Apple, Microsoft/Outlook, Facebook) + enterprise SSO; enforced MFA for staff/admins.
Primary database	Azure SQL Database	Stores all institution data; Row-Level Security keeps tenants isolated; built-in HA.
File storage	Azure Blob Storage	Documents/media; a write-once container holds the tamper-evident audit log .
Secrets & keys	Azure Key Vault	Central, encrypted store for secrets and per-tenant encryption keys.
Messaging	Azure Communication	Email, SMS and one-time

Component	Azure service	Role / why it's here
	Services	passcodes (e.g. enrolment confirmations).
Monitoring	App Insights + Log Analytics	End-to-end tracing, dashboards, alerting, and long-term audit retention.
Security posture	Microsoft Defender for Cloud	Continuous threat detection across app, database and storage.
Backup	Azure SQL PITR	Automated point-in-time recovery.

5.3 Component briefs — Phase 2 (activated as modules launch)

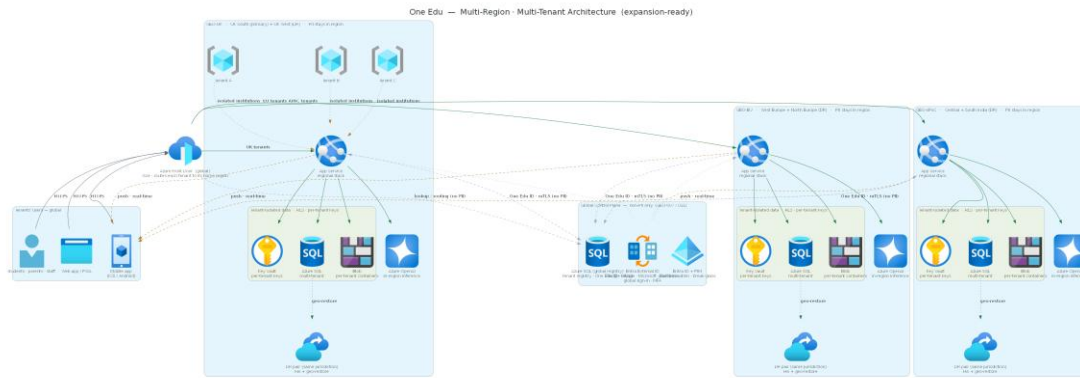
Component	Azure service	Activated for
Real-time messaging	SignalR / Web PubSub	In-app chat and live updates.
Cache	Azure Cache for Redis	Sessions and rate-limiting at scale.
Async jobs / events	Azure Service Bus	Background processing and data migration (e.g. from Moodle).
Mobile push	Azure Notification Hubs	Push notifications to iOS/Android (via APNS/FCM).
AI	Azure OpenAI + AI Search	In-region AI insights and tutor (retrieval over the institution's own content).
Public/partner API	Azure API Management	Versioned external API, webhooks, rate-limiting.
Network hardening	Private Endpoints + NAT	Private-only data tier for the strictest compliance posture.

5.4 External services (not Azure)

Service	Purpose
Payment gateway (Stripe / PayHere)	Card capture & PCI-DSS scope; One Edu never stores card data.
Legacy LMS (Moodle)	Source system for data migration into One Edu.
Apple / Google app stores	Distribution of the native mobile apps.

6. Multi-Region (Multi-Tenant) Architecture — Expansion Model

As One Edu grows into new countries, the **same regional blueprint is replicated** per geography. This is the target operating model and demonstrates expansion readiness to investors and regulators.



Multi-region, multi-tenant hosting architecture for One Edu

Figure 2 — Multi-region, multi-tenant architecture. Each geo-region is a self-contained data boundary; a small global control plane (non-PII only) routes tenants and manages cross-region identity.

6.1 Key concepts

- **Global front door & routing.** A single global Azure Front Door receives all users and routes each institution (tenant) to **its home region**.
- **Global control plane (non-PII only).** A small **Azure SQL global registry** holds only non-personal metadata — which tenant lives in which region, the platform-wide student identifier (“One Edu ID”), and billing. **No student personal data is stored globally.** Global identity (Entra External ID) and platform-admin access (with just-in-time elevation) also sit here.
- **Self-contained regional stacks.** Each region (e.g. **UK, EU, APAC**) runs the full Phase 1/2 stack — App Service, Azure SQL, Blob, Key Vault, AI — and keeps **all personal data inside that region**.
- **Tenant isolation.** Within each region, multiple institutions share the stack but are strictly isolated (Row-Level Security + per-tenant keys). One institution can never see another’s data.
- **Disaster recovery.** Each region has a paired DR region **within the same legal jurisdiction** (e.g. UK South ↔ UK West).

6.2 Why this matters to the Board

Benefit	Explanation
Data residency / legal compliance	Each country’s data stays in-country — meeting GDPR, UK GDPR, and equivalent laws.
Trust & sales	Institutions and regulators can be shown exactly where their data lives.
Scalability	New markets are entered by stamping the blueprint — no rebuild.
Resilience	A regional incident is contained; other regions are unaffected.
Controlled cost	Most shared services are amortised across tenants, so per-tenant cost falls as the platform grows.

Note: the global registry uses **Azure SQL** (consistent with the rest of the estate). A globally-distributed database (e.g. Cosmos DB) would only be considered if very high multi-region write throughput is needed in future — it is not required at launch or early growth.

7. Security & Governance / Compliance

Security and data-protection are **engineered into the hosting platform**, not added afterwards. One Edu processes the personal data of **children** across multiple jurisdictions, so the design

assumes the highest bar. This section briefs *how the components deliver* security (defence-in-depth) and governance (GDPR-class compliance).

7.1 Security — defence in depth, layer by layer

Every request passes through multiple independent controls, so no single failure exposes data.

Layer	Control	Delivered by
Edge	TLS encryption, Web Application Firewall (OWASP rules), volumetric (DDoS) protection	Azure Front Door + WAF
Identity	Social sign-in (Google, Apple, Microsoft/Outlook, Facebook) + enterprise SSO; multi-factor authentication for staff/admins; conditional access; just-in-time (“break-glass”) privileged access	Microsoft Entra External ID + PIM
Application	Server-side authorisation (role-based), input validation, per-module entitlement enforcement	Azure App Service
Data — isolation	Per-tenant Row-Level Security ; no query path can return another tenant’s rows	Azure SQL Database
Data — encryption	TLS in transit; AES-256 at rest; sensitive fields encrypted with per-tenant keys	Azure SQL + Blob Storage + Key Vault
Secrets	No credentials in code; centralised, access-controlled, automatically rotated	Azure Key Vault
Network	Private-only data tier — no public database endpoint (Phase 2 hardening)	Private Endpoints + Virtual Network
Detection & response	Threat detection, anomalous-access alerts, vulnerability findings	Microsoft Defender for Cloud
Assurance	Independent penetration test before launch; OWASP Top 10 clearance each release	Process / gate

7.2 Governance & data-protection compliance (GDPR-class)

Each regulatory obligation is mapped to the hosting components that satisfy it — so compliance is demonstrable, not aspirational.

Obligation (GDPR-class)	How the platform meets it	Component(s)
Data residency — keep data in-country	A tenant's personal data lives only in its geo-region; the global plane holds non-PII metadata only	Per-region stacks; Azure SQL global registry (non-PII)
Confidentiality / tenant isolation	Row-Level Security + per-tenant encryption keys make cross-tenant access impossible	Azure SQL, Key Vault
Audit & accountability	Tamper-evident, write-once log of security-significant events (logins, exports, admin changes)	Blob (immutable container) + Log Analytics
Data-subject rights (access / erasure / portability)	Records assembled across all modules for a subject; export and erasure workflows with lawful-retention overrides	App Service + Azure SQL / Blob
Lawful basis & consent	Consent and legal basis recorded per processing activity; withdrawal triggers downstream workflows	Application data model
Retention & minimisation	Automated purge of records past their retention period; statutory floors enforced	Scheduled jobs + Azure SQL
High-risk processing (DPIA)	Biometric/AI features are blocked until an approved Data Protection Impact Assessment exists	Application governance module
Breach response	Anomaly detection → containment → templated regulator / data-subject notification within the statutory window (e.g. 72h)	Defender for Cloud + process
Transparency	Per-tenant data-residency attestation; AI usage notice to guardians	Control plane + application
Children's data	Maximum-privacy defaults,	Application + identity

Obligation (GDPR-class)	How the platform meets it	Component(s)
	guardian (or adult-student) consent, no harmful profiling	

7.3 Standards & frameworks

The platform is designed to support, per geo-region: **EU GDPR, UK GDPR & DPA 2018, FERPA, COPPA, PIPEDA, India DPDP, PDPA (Sri Lanka / Singapore / Thailand) and UAE PDPL**, plus UK safeguarding guidance (**KCSIE**) and the **ICO Children's Code**. Targeted platform certifications: **ISO 27001** and **SOC 2 Type II** (and **Cyber Essentials Plus** for the UK).

In one line for the Board: data is encrypted, isolated per institution, kept in its own country, fully audited, and every privacy right (access, erasure, consent, breach notification) is supported by a specific component — making One Edu *demonstrably* compliant, not just compliant on paper.

8. Availability, DR, Scalability & Operations

Aspect	Design
High availability	App Service runs ≥ 2 zone-redundant instances; Azure SQL has built-in in-region HA (99.99%).
Disaster recovery	DR region within the same jurisdiction; SQL geo-restore; recovery objectives RTO ≤ 15 min / RPO ≤ 5 min.
Backups	Automated point-in-time recovery (up to 35 days).
Scalability	App Service auto-scales; database scales by tier/serverless; architecture supports thousands of tenants.
Observability	Structured logging, distributed tracing, dashboards and alerting (App Insights / Log Analytics / Grafana).
Environments	DEV, UAT and PROD (Section 9).

9. Environments & Indicative Cost

Three environments are maintained, sized to the lowest viable SKUs to control cost. DEV mirrors UAT on the application + database tiers for parity.

Environment	Purpose	Indicative monthly (USD)
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Environment	Purpose	Indicative monthly (USD)
DEV	Active development	~\$103
UAT	Testing / client demos	~\$169
PROD (with HA)	Live institution	~\$672
Development tools & SaaS	Team of 5 (incl. Claude, Azure DevOps, app-store licences)	~\$150
Initial monthly run (excl. manpower)		≈ \$1,094
Phase 2 add-ons (on activation)		+ ~\$565

*Full breakdown, assumptions and GBP figures are in the companion workbook **CostEstimate-OneEdu-Board.xlsx**. Figures exclude staff costs and variable payment-gateway fees.*

10. Delivery Roadmap (Phasing)

Phase	Scope	Hosting impact
Phase 1 — Launch	First institution, single region, core modules	App Service + Azure SQL + edge/security/monitoring (~\$1,094/mo all-in).
Phase 2 — Module growth	Real-time, AI, public API, push, hardening	Activate the amber-zone services as each module ships (~\$565/mo).
Phase 3 — Geographic expansion	New countries/regions	Replicate the regional blueprint; add the global control plane.

11. Assumptions & Risks

Assumptions - Single institution (~5,000 users) at launch; UK South region; Windows on App Service. - Email handled by a separate provider; payment handled by a PCI-DSS-certified gateway. - Costs are list-price estimates; final figures via Azure Pricing Calculator and vendor quotes.

Key risks & mitigations - *Demand growth beyond sizing* → tiers scale on demand; reserved-instance pricing reduces cost. - *Regulatory change per market* → per-region data boundaries and configurable compliance profile. - *Third-party dependency (payments, push)* → standards-based integrations; replaceable providers.

12. Glossary

Term	Meaning
PaaS	Platform-as-a-Service — managed cloud services (less to operate than servers).
WAF	Web Application Firewall — blocks common web attacks.
HA / DR	High Availability / Disaster Recovery.
RTO / RPO	Recovery Time / Point Objective — max acceptable downtime / data loss.
Tenant	A single subscribing institution and its isolated data.
Row-Level Security (RLS)	Database feature enforcing that each tenant sees only its own rows.
Geo-region	A compliance/data-residency boundary mapping to Azure regions.
One Edu ID	Platform-wide unique student identifier, consistent across institutions.
MFA	Multi-Factor Authentication.
PITR	Point-In-Time Recovery (database restore).

Prepared for board review. Diagrams are available at full resolution in the project's docs/images/ folder; detailed costs in CostEstimate-OneEdu-Board.xlsx.